

ActInf Textbook Group ~ Cohort 4 ~ Meeting 8 (Chapter 3 part 2)

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okay it's cohort four and July 25th and
we're the second discussion of chapter
three

so before we look at the questions is
there anything that like anyone just
wants to start anything about the
reading or anything
even maybe some recent uh
live stream or or course or anything or
just

what their reading is like now of
chapter three

yeah go for it

so I I was thinking about this idea of
synchronization between the inner model
and the outer model which another term
for that could be adaptation to the
environment

active adaptation modifying the
environment and building a model of the
environment but Bronwyn mentioned
personality disorders and one of the uh
features there these are very fuzzy
Constructors being stuck in maladaptive
patterns that don't really fit the

environment and person is unable to
change them thus their energies uh they
swim in free energy so to speak and they
suffer and there's negative effect that
is present all the time I I do think
that these models are a little too
straight and narrow to account for
Psychopathology and I do think trying to
explain something like that with sort of
a a need formula is is just not ready

yet I mean I I think that these are not doing it justice because you know uh people exist who spend decades in maladaptive behavior and they don't uh do much about it to get better so I I just find it that you know when when we dictated and say this is the law this is the first principle and this is how the living thing does it bacteria perhaps but in a person with like malignant narcissism no they don't

yeah it's a really good uh

question like what is the role of the math in the science what's the role of the science

quote science what's the role of the math in the science

how just broadly extrapolative is it in its interpretation

um and then what's the role of just any any how is how is a math and science mobilized

with personality

what does a principle matter in the specific

what does a specific matter for a principal

those are all huge questions

I think you'd be interested in a cybernetic Big Five Theory they have a few papers where they connected to free energy principle to try to go at different psychopathologies and try to see how you can like think of in a cybernetic perspective it might be something you might be interested in I'm familiar with the big five Theory

it's also statistics based I'm a
clinical psychologist and I think it's
uh has nothing to do with personality
disorders I think it's again trying to
uh uh quantize
but but still you know we're going for
mathematics trying to Define I mean it
would be good to to bring somebody in
the room who has a borderline
personality disorder and try to apply
that theory onto them they're
predictably unpredictable there's
something missing like I don't know
field Theory or Chaos Theory framework
it's just it it it uh it doesn't match
my empirical data of interacting with
his patients on a daily basis you know
and it doesn't come from papers it comes
from seeing what they do in life over
and over and over again it doesn't work
they still do it so uh
yeah one way I I I might kind of ask
this would be the personality traits are
more about the
um characteristics of the paths in a
cognitive model so it's like certain
type of Dynamics
in the map not the territory then in the
you know maybe a clinical setting first
off it's engagement directly rather than
engagement with the model
and also there's perhaps less of a focus
on just like
um in the in the short term about just
um population scale patterns and maybe
more emphasis on a person's action
patterns
and so then how can these two

sides kind of support each other in an

applied cognitive science

rather than like disengaging

[Music]

characteristics and uh cbtf cbft

uh and cybernetic Big Five Theory as

like another layered uh outside of just

traits

what was the name of the thing outside

of traits uh characteristic adaptations

oh yeah also anyone of course can always

be just clicking on the cursor and

following and taking notes

like people can have things just it's

all good if people want to to discuss or

just filling in other angles

but thanks everybody who has shared

any other

thoughts before we look at any of the

questions there could be anything

um

actually yeah it just as a brief

follow-up to the preceding conversation

I was just curious if anyone

finds interest in the notion of like

hyper precise priors I see that come up

very frequently in uh applications of

active inference to Psychopathology and

um I think basically the idea that

there's a like a very strong belief that

a certain model is correct to where it

kind of skews and coming

um you know sensory information and it

kind of continues

um to to impact future actions

um despite like disconfirmatory sensory

evidence I mean I I'm just in response

to the earlier conversation I found I've

found that to be really interesting and I thought it dovetailed really nicely with with studying Psychopathology actually yeah I I think it matches well to something like delusion because if I believe firmly that neighbor walks in my house and changes the furniture when I am outside then and it doesn't get uh updated based on me calling the police and calling you know installing cameras then that sort of hyper precise priors exactly but I I do think that again trying to stretch these approach to different kinds of Psychopathology like as I've mentioned say personality disorders right so a lot of personality disorders are uh related to early experiences where the child or adolescent try to adapt to the environment which is very difficult to survive it emotionally and they've developed some habits that were not solutions back then but there's nothing better that was available to them and they carried it through into adulthood so uh for example you know a child often in a divorce of parents ends up thinking that he is the reason why they're divorcing they end up internalizing distress and saying I'm the guilty party and then they get in trouble because if I'm in trouble then maybe these two Knuckleheads will get together and then you see an adult patient Psychotherapy in their 30s that apologize every session for everything you know and

that's an example kind of and it's not necessarily I mean it has a completely different flavor from delusion and I don't think the language is hyper precise priority is sufficient to distinguish between the two there's qualitative differences uh and so I guess that more stuff is needed to describe the difference of it what about something like uh obsessed impulsive disorder uh where you could think of it as a from like an ecological psychological perspective as uh a low tolerance for affordances so yeah low tolerance for uncertainty basically

um

that that would not do it justice that would be a very simplistic approach to OCD because there's different subtypes people who do washing and check-in are very different from borders

so

uh uh you could have people dysregulated and people were just regulated childhood OC there's a lot of it all together this is an example of trying to you know uh fit something very heterogeneous complex into a simple formula and just saying oh we got this cool Theory let's just explain OCD with it no I mean I think if we want to do the OCD Justice we need to start with a lit review go to PubMed and read all the theories about OCD genetic influence you know environmental influence uh you know chemistry everything else put it all together and then we should do it so this approach

from first principles to something as
complex as

uh it's almost Farr is a psychiatrist

uh Colonel Friston is a psychiatrist a

lot of people who aren't they want to

just you know schlep a formula to OCD

and it just it doesn't work

well again what what is really the role

of a of a specific model

uh

construction of those models is usually

not uh utilized as a full account

but maybe in it it can seem that way or

be used that way in certain settings

like to guide action

um

and then also like

one good thing about generative modeling

is that it allows

really specific discussion of what has

and hasn't been specifically modeled so

first principle is just because

something feels like consilience in

one's own mind does not equate to an

explanation let alone useful in the

world

so

then there can be a discussion of like

okay what um sensory modalities and

interoceptive modalities are taking into

account and what how's childhood you

know in this specific model of these

people who have OCDs like oh there's no

generative model well we have chapter

six we have chapter nine

you can build the model but definitely

if it's just speculation about what

inclusion or exclusion of variables

you're just like kind of casually
curious about or professionally curious
about it's all speculative
so I think that kind of supports a more
bright line because then people can just
spitball a little more broadly
structurally
which everyone's gonna have some
expertise in some area but then also
it's clear like when someone's not
referencing a specific generative model
it's just a pure hypothetical so how
would uh what what um what would a
hypothetical linear regression account
for nothing
so what would a hypothetical cognitive
model account for well maybe something
but doesn't have to
so I have a question is this more of an
issue with how we're classifying
psychopathologies or is this as in like
each classification is too broad perhaps
and we need to be more specific maybe
and because maybe like the same like you
were saying like there's many different
types of obsessive-compulsive disorder
so maybe the ones that pertain to uh a
low tolerance for uncertainty
would need its own specific
classification
versus other ones that are different
yeah
which is a question there
there's many issues there one is you
know we apply mathematical apparatus to
another field we're trying to either or
cybernetic theories or something else
and it doesn't always work for example

in Psychology there's something called ambivalence which is hating and loving the person at the same time that concept violates the logic of Aristotle and all of mathematics is based on Aristotle and logic so there you go you can't use the absolutely fundamental framework while you're trying to use something more sophisticated like classifications and whatnot and there's many critiques of DSM classifications as the declassifications but the important thing is the construct is complex and then you know before trying to model it or explain it we should study the existing literature and and go from there because otherwise we are trying to model our fantasy of OCD and what sort of the the Practical reality of OCD that is you know like uh sadly it's it's uh still very bad like schizophrenia we know about 14 possible causal factors but there is no coherent fear of schizophrenia that everybody exists on nor depression it's the oldest condition in the history of medicine but there is no integrated theory of depression still it doesn't exist and so this active inference is one of the attempts to do that but it still falls short like if you use active inference and you try to explain everything from autism to I don't know motor disorders it doesn't fit it doesn't capture the the core of autism it just it it uh it seems to fit you have this resonance you're like oh that's cool I think I've

explained it but if you actually see people with Autism you see that you're explaining something else I mean because the core of autism is mentalization deficits so uh I

I don't know if it's a good answer but it's a it's a very messy field it's very different mathematics is very kind of you either proven theorem it's either true or false

in political psychics and there are different things

yeah

that's why uh modeling work is part of an integrated

scientifically plural framework

I need to do both oh just just it's it's

a suit it's a super over fit

like it's always about reading a monks in between the citations and everything

else it's not about one paper one experiment one data set it's always

about being explicit about the uh

interpretation and how the researchers

model was made and so if somebody shows up talking to you

um with a lot of personal experience and

or a lot of meta-analysis I'm sure

that's a productive conversation

also can be productive

um someone who might not be aware of

some of those complexities but yeah

that's also there's also just like these

are it's always like active inference or

is this about just like any kind of

modeling

numerical or statistical modeling

because someone could also make I I

believe in identical we're not linear
regressions our health records aren't
linear regressions that's not the body
that's not decision making it's just
linear regression so Decades of
insurance being calculated that way for
example so if that ends up using
cognitive modeling that that's kind of
the gray Zone slash question
is what is that kind of cognitive
digital twin
not just the accumulation of information
from a privacy perspective
but that kind of like more decision
making
representation which of course is still
just map and not territory
but that's a different area
just one kind of modern connection I
don't know do people have any thoughts
on that
or how this relates to
um different machine learning like
how does it relate to machine learning
or what's happening right now with with
synthetic intelligence
um well
I know that uh
but it seems like self-attention
mechanisms are also trying to minimize
perplexity and there was this recent
paper that they're using called like
evolutionary Transformer architectures
where uh they basically add like an
evolutionary block to it which allows it
to so you have multiple sort of models
and they're also all trying to minimize
perplexity

um

and then I know like GPT the latest with
GPT was they it was like leaked that
they're using

um

they're using a

it's called a mixture of experts or
yeah basically like a similar type of
algorithms to just find what comes up
with the best sort of output among many
different models so I just thought
that's kind of similar to like some
things that were mentioned in active
inference especially this chapter
yeah that's one great connection also
with Collective behavior and evolution
and active but also attention I know
Holly and uh others have have kind of
thought about this or or considered a
little bit like attention

mechanisms first off attention is just
one of a huge variety of cognitive
mechanisms

anticipation expectation attention

so

um this is we can think about the
regimes of attention which is a concept
from the cultural work

with in cultured patterns regimes of
attention

uh to sensory into different things and

so

there could be a Transformer with a real
cognitive model

explicit semantically interpretable

active

figure 4.3 type cognitive model here's
the explicit attention function

for example and that and that there's a lot more say in that area

but that's for those who are interested

I feel like that's a cool you know thing to look into

any other chapter three thought or we can go to the questions

all right everyone can vote or just uh

also open the question

all right

note that the resulting preferred course of action or policy is directly

measurable in experimental settings

whereas a value function or prior belief needs to be inferred and hence is more indirect if not tautological measure

I'm not sure how or why the measure might be tautological in the case where the value function needs to be inferred what is the tautology here

anyone have a thought on that

all right well I I I am not exactly sure

but

we can contrast reward learning with active inference type generative modeling so reward learning you're going

to assign some utility or value to

different states and then pursue

rewarding States like different chess

pieces get a value and then you want to

maximize that expected value

that's the imperative for selecting

different actions

then inactive

action is going to be selected based

upon it being the most

uh basically

it depends on the kind of agent because
different agents have different kind of
relationship with the future they're
simple agents or just
um kind of inert
or have deterministic action policy but
then there's the whole open-endedness of
cognitive systems so
um
they act to learn but they also act to
realize their preferred uh their
preferences about sensation
but however you look at their behavior
whatever you think about their behavior
whatever you propose as as guiding their
behavior
it is what is directly measurable in
experimental settings
which is shown in figure 9.2
oh 9.1
which we talked about actually in the um
other cohort earlier today
so we're talking about what we present
as stimuli and what we observe as
Behavior
so that's what this modeling approach is
Tethered upon that
setting where it's being a cognitive
model is being projected
in contrast if you propose some second
order function about the behavior like
well it likes being in the light not
because it expects to be the light and
that's the most likely thing but because
light is rewarding and so then it's
going to pursue reward
you've basically proposed
an ad hoc

potentially principled
but at the very least indirect
and possibly just self
describing
something that isn't this though
maybe there's some specific case but it
seems pretty vague
I mean anyone else think about the
tautology or
I sort of uh
just took that that quote as referring
to like uh so preferences
pardon me I'm still to kind of trying to
wrap my head around this like like
you're already starting with like a set
of preferences I suppose that you're c
um Matrix right and um
so there's the tautological aspect is
that I suppose whenever you're modeling
you always have to start
with some kind of C like we it's it's as
if it's saying that active inference Is
Not An Origin story it's just like we're
already there and so like there's no
explanation for the starting point we
can only measure what maybe results
following after our priors are already
defined something along those lines
nice that's a great thought
um
what there's a few ways to look at it
one one on the more formal side
is like
if we think about things that don't plan
they're going to be amongst other things
like that just the same and all like
diffusing particles
so like they have no preferred state in

the gas box

um so they're they're like they're

behaving like just kind of like

classically from experimental

perspective or like according to

thermodynamic equilibrium whereas

systems that have an attracting set

homeostasis

or

um acting with the c to kind of anchor

the attracting set

without any pragmatic value policy

selection there's just policies are

driven by pure epistemic which in

abstracted settings maybe on optimal

situation

but for embodied settings that's not

the general principle

so that's why there's those two

pragmatic and epistemic

which helps

um

give a lot of degrees of freedom because

this is a value that's calculated for

the specific model that somebody

constructs like in chapter six so it

can't be just generically said like well

the free energy of an estimate goes down

when it gets a seed it's like

it's it's way way speculative to say

before having like the specific kind of

model that's even being discussed

so and it's just it's explicitly what is

being

um the attracting set itself

not a proposed second order

variable

just one possibility

but under another approach is like that
is like backs like is it about modeling
or is it about active inferences like
okay what other what other approach
can be taken in this setting is this not
a worthy setting or like is there
another way to approach this
and what would that what would that kind
of a variable or what would that model
be

so it doesn't have to be all only every

[Music]

model

but that's kind of the the golden Hammer
Golden Nail or whatever

um

now that can be a simplifying

um

cognitive heuristic

which has all the benefits and
trade-offs of any cognitive system
engaging in a similar heuristic

but we're in the textbook group so we
can like learn and like scrutinize it or
like use it in ways that that don't
necessarily make specific sense in any
given one person's
experience of using it

leaf blower

any other thoughts on this question
what are the minimal contents of the
model encoded within the internal states
of an activation

I'm presuming that

one a statistical model of the hidden
causes of a sensory data and two
preferences about the state the
generative model needs to be in order to

ontologically persist Through Time

I feel like one is specified in the
section but where in the generative
model are the preferred States located
who has any thought on this
so would this not be what they describe
as like the motion of the action States
as the
prediction of the preferred States as
well

I don't know if that's
that covers that aspect

okay

possibly

um

like in the actual person I think it

kind of

can you say the relationship again

yeah so in the book they describe uh how

the Markov blanket not only is

predicting uh

sensory straight States it's also

um

doing so for the action States as well

I don't know if I'm phrasing that

correctly

but

if I can find it

one equation based way to to look at

this

is so where in the generative model are

preferred States located

there the sea here

so they're used in the free expected

free energy calculation so if if we're

talking about a kind of of simple model

not a simple system but a simple model

where the cognitive states are just like

a pass-through or you know a divide by two or you know there or it just is just a noise so the action policy in the sensor are uncorrelated or uncaused or anything like that like those are all special cases but action policy selectors which might be one step selectors like instantaneous selectors doesn't have to be an explicit planner per se

um could be like a gradient based system with bacteria

um pragmatic value is where preferences are loaded

and pragmatic value rebates policies accordingly to how

close you get

to the preference Vector slash variable and how relevant that is in the behavior is situational there's no generic answer because the other component is about epistemic gain

and so this is the the relationships that's described in all those special cases where this or that doesn't apply so it's a totally good question

but it's a very you know

this is one of the key differences between reward learning and active inference in active inference the preferences are explicit over sensory

States that's a relationship with perceptual control theory

and it prevents overfitting because it's about the incoming sensory data

predictive processing all of this

um there's no proposition of a reward or utility function

what the agent is supposed to be doing

what our inference does procedurally

is just

these optimizations on the specific

proposed generative models

so first principle may do last work

generically

principle of least action people are not

always bringing this out to talk about

well this is why this plane did or

didn't fly

well this one was just following its

path of least action

so that's not an explanation or an

account

and I think that that's that code in

what you shared Lexi like sort of about

like about uh particulars and generics

and everything like that

so

first principle less specific work done

by the framework understanding the

framework

very prerequisite for modeling

understanding claims for for of course

just the joy and the fun everything like

that all the fun we have

um but understanding claims

like evaluating whether a given um usage

of the active inference ontology is like

valid

but then to do the applied modeling and

to understand it in the way like that a

graduate student type

might

then these are only kind of the enabling

pieces kind of like the L2

least um some squares minimization has a

relationship with linear modeling
but a more statistical student might
learn more about the underlying basis of
the linear regression
more empirical statistics student might
just apply data set modeling not go too
deep into the theory
just want to kind of provide some of
that context because like Kristen at all
have explicitly said this is for this
textbook that that were
gathering around like that it's a kind
of Masters student
again taped out for what it means and
there's always more variability like
within them between all these other
things but Master students
getting on the way to doing modeling
in this approach
based upon
reading the epistemic first half of the
book
I don't know what piece of course we're
having fun figuring it out
and then going into the modeling in
parallel
following along after the chapter six
with one's own kind of
specifications or just like Explorations
for different systems
some stubs of which we have and we can
collaboratively do too
but it's kind of just a general comment
because like if the philosophy is is
amazing it really is and people bring so
much to these discussions that that I'm
just so glad that that we have them and
everything like that

and a lot of it is is very
uh General and importance outside of
just active inference
which is really like uh in in the in the
textbook sense
like uh
comparatively
can be narrow
because it can feel like there's like a
lot of like things to get like rabbit
holes to get lost in and everything
Crags to fall into
but
I I I hope it throws a little context
hence just like regularly working with
material and just broader sketches
natural language that is consistent with
the equations but you know there's only
so much mystery in one's time in in life
with a mystery with the equations Alexi
yeah um thank you so much I mean I think
you described it beautifully I just
wanted to also comment that everything
we're talking about only applies for
systems just like when it says early in
the chapter that have an attracting set
this is fundamental quality of the
systems you describe on this figure 3-3
when they show a system that doesn't
have an attraction the system that does
have an attractor active versus
inference models talk about the ones
that do have an attraction okay kind of
converge right
uh and so I recently asked Carl via
email and he asked lanceland Acosta as
well what if they're attracting set
changes

for example if I sleep there's one
attraction set when I wake up there's
another tractor set and my entire system
bifurcates it evolves from let's say
periodic attractor to a strange
attraction

and they are working on that right now I
think this is a work in progress but
they will create a sort of a
hierarchical structure where the
attracting set itself is not kind of set
where you can describe and say that's
the attraction set but it can actually
change in and of itself and that's
that's a new interesting development
that may uh

that may help I don't know how people
feel about that
yeah that's awesome that sounds really
cool one one piece to bring in here this
is Axel Constance like

extremely

um kind of uh girdle Escher Bach
type

uh description of fep which kind of just
had to be said

but he he laid it down

and

the simple point I'm making this paper
is that the free energy principle is not
concerned with the sufficient conditions
of existence

but rather

with what must have been the case given
that you exist it's not about

figuring out what it takes to be alive

it's about figuring out what took you
there

that's just so deep and I think it it
gives a difference
accounts for a lot of settings just like
the surprisal versus reward
does
that um
now that that in place I don't know
exactly how they're related here because
maybe this could be taken you know but
it's like saying we're making a
measurement theory for measurable
objects
we have the blur so we know when there's
no signal that's just noise
that's going to be because that's the
Baseline informationally
and then we're going to have a theory of
measurements an analytic framework for
measurements cognitive measurements
observables and unobservables
um
a four measurable systems
or if you do a hypothetical it doesn't
have to be like an Adaptive system
so that's what's fun in the textbook
group is like we can imagine just like
okay let's just imagine that the
external state is just
alternating between two states
it's just like I mean it's great to
really wonder about these um really
contemporary examples and everything
like that but like
pedagogically
a lot of these existentials are
non-relevant it's good to keep them in
mind because I think a lot of the the
technical part like people can study and

um and the math learning group and
hopefully like it's you know
there could be good math discussions but
this is like I'll bring so much of the
broader context in
what everyone shares but because the the
math is not a metaphysic
and this
inversion of
um
of uh conditioning
like most people if they think about
um a theory of everything they're going
to think about it like from a sort of
material up
like how things came to be
versus
um so versus
at least
the measuring or modeling framework for
Signal versus noise
which does a less slash different work
per se than the than the um the accounts
of how something like comes to exist
because different mechanisms for things
coming to exist
they might have from it for a given
generative model
like they might have similar or
different Dynamics
but the model is not gonna like I think
here to um JF classier at The Institute
in his work with the uh symbolic active
inference
robotics
and about how we can have different
implementation architectures that can be
modeled

from a behavior like from an active
inference Behavior Uh perspective
but this doesn't mean that everything
has to be like running this exact
statistical model as a kernel
that's like figure 4.3 is like super
schematic
extremely it's like it's it's not a
claim about a system it's just super
evocative about the kind of structure
that can be expressed
for in the most minimal
way

and then there's a lot there's a lot
more to do and everything but
that's why it gets kept it keeps on
getting returned to
because it helps give a first pass
Separation on like priors
sends to Hidden as hidden state to
sensing

World model

and action model

so these are some of the essential
features of of

as it turns out a nerd and sentient
creatures

but this is not like

yeah

I hope that's not too too contextual but
just want to clarify because I I can I
can imagine slash C that that
what the equations do is a primary
concern slash awareness
and uh so so I think that it's very
um worthwhile for people who want to
continue like learning and thinking
about it like

just

um sharing their perspectives on

what these maths actually do

because then it's just parallel streams

where one is just like like

now I don't know just what do people

think about that or or how do we really

connect up even just in these short

discussions

some of these mathematical details which

are kind of just like

just defined as true essentially

and all these other systems of Interest

since like ways of knowing that we all

are interested in

um

I hope I'm not too like far afield from

the way that things are being framed a

moment ago but um I mean as far as

something like uh simultaneously or

simultaneous like attractor sets that

might be competing with one another for

example

um and the idea of like you know you

have a series of preferences

um I've I've not had too much trouble

wrapping my mind around the idea that

you can have you know competing uh

tractor States or competing preferences

if you need to use the restroom and

you're hungry and you're very sleepy all

at the same time and so we have these

models that you know whether it be the

discrete version or the continuous

version they all happen in time and they

all involve like policy selection over

time so for example if I decide oh like

you know it's I'm very tired but it's

only 2 p.m I could take a short nap and then I could get up and you know have a snack and use the restroom whatever I need to do but you can plan things out over time and and as far as the relationship between that and the math goes

um like we have that nice like expected free energy equals you know epistemic value the real you know the combination of the epistemic value and the pragmatic value and so your C's your preferences are only in

um pragmatic value as you mentioned earlier Daniel

so whenever you aren't sure what to do you have these competing preferences like there's a almost like a uniform probability between them like which one is going to win out I'm not sure you would rely upon doing actions that involve gaining more epistemic value like oh well I'm going to have to spend a moment like planning and what makes the most sense here can I come up with a plan that helps me to like resolve some of this uncertainty going forward and do that and you know um or maybe you have a certain belief that one of these things is just incredibly important and so you you stick with that one in the first place or maybe you become conflicted not to not to backtrack but on the topic of Psychopathology I'd be interested to see someone just attempting to model decision making and planning and so on in the context of someone who might have you know some kind of anxiety disorder

for example or in fact like maybe you have an additional Merchant behavior that is spending you know five to ten minutes or maybe a few hours like ruminating having issues about what to decide upon I read up I'll try to find it but I find an interesting paper that tries to get at you know modeling uh rumination in that kind of context so that those are all my comments here on the idea of like competing preferences and relationship to the mathematics oh yeah

nice thank you

one other

um aspect uh to kind of so whether it's the visual form which is describing the sparsity of the base graph like which nodes

connect to wish knows this is this is kind of like the skeleton of the model generative model this is where the time concept is embedded this is where the action concept is embedded sense making concept prior concept

so it could be viewed this way and then the equations are describing like statistical operations on that and and that's why uh

but

whatever structures proposed it to be applicable to any actual system it's a parameterized data set

specific model specific data set like saying let's um imagine that uh childhood just imagine the childhood

experience one two and three are associated with um with cybernetic B5 or DSM such such and such just imagine that they're that case so what do you think about this being the case this one's associated with that it's like well I don't know you haven't specified the model or applied it to a data set but it's like oh yeah but you can like see that that would be a really important thing right it's like okay perhaps but that's a purely verbal argument that's what I was referring to earlier as just speculative that can be an intuition pump and I believe serve a lot of other functions and and communication functions and other features but um that is not what a scientific account consists of which is a much slower term um pluralistic um synthesis not not to be like too old school on that topic but I think it helps enable open-ended discourse which I believe is important and when people watch so many different backgrounds and and and familiarities that that of course everyone's going to be like thinking and saying things that we would not think and say because we come from so many backgrounds but just what happens in a discussion around a textbook is not um is not it's in a Continuum of communication but it's not of the type of a scientific account

ajith

yeah so

um I have a question about something
else but

uh just kind of the idea that like our
behavior is governed by a principle of
least action

so how does this affect like

um assumptions of like measurement
Independence so like some experiments
uh for example like the one they use to
sort of

even tests so I'm not my background
isn't physics so

um if anyone knows correct me if I'm
wrong but to test like

uh Bell's inequality there's an
assumption of measurement Independence
that like you can actually make
decisions and it's not actually affected
by sort of the environment or what's
going on uh and this is sort of a way to
actually like

make it so you can actually test certain
things and if you can assume measurement
Independence that kind of breaks science
there or a lot of people think it breaks
sort of science there so uh Concepts
like super determinism uh

is one that assumes that every sort of
like Behavior or action is something
that uh kind of is dependent on sort of
all the past States as well basically so
um yeah I was just wondering like I I
don't know exactly how like active
inference would play in that like does
this go against measurement Independence
if it is a principle of least action

because it's like going in the direction
like the most least effort basically
yeah there's a ton of areas I'm not
familiar with every single aspect there
this is as Toby Smith
um uh Sinclair Smith uh called it in his
recent dissertation which was uh live
stream 54 series
he called it
compositional cognitive cartography
we're talking about maps variational
free energy expected for energies are
properties of maps that are human
constructed and modeled hashtag John
Dewey all of this other pragmatism and
and threads that are brought in there's
nothing generic to say about free energy
or expected free energy except for the
truly generic active ontology type
Expressions so that's why we can like
open up and have a lot of richness with
the chapter six type discussion about
modeling because they're so far to go as
people are encouraged to do in the
textbook group
um and through time with just like
whether you're coding is that um this or
that I hope we can develop better uh
coding and all these other things
collaboratively anyone's welcome to do
this
a lot of these are questions about
specific model settings and um
there's a big Continuum between kind of
touching the theory and touching the
system
but that's the inclusion that we get to
cultivate with the right Niche if we can

create the right artifacts that support
people who are different understandings
because like there's a lot of threads
that were brought in relatively
um

early in terms of like the connection to
ecological psychology
and predictive coding some of these
things are only finding any
empirical uh um support but with rapidly
expanding amounts since like 1999 with
kind of the classic round Ballard type
predictive coding
models

so not there any given communities like
acceptance of something matters or what
is in the literature because there's
kind of like the timelessness of the
ideas in a sense but
um

like

again I just there's a broad range
between the mathematical and the
engineering and all these other
perspectives

so it's very wide that's what makes it
wide open in terms of like creating like
well what about accounting from this
perspective

that's those are almost always good
directions to go if you're you know here
and curious

and uh what What's so fun is as people
continue to go deeper in their um work
that it it can go from like kind of
um

uh tautologies

um or or just just um just active

infantology

scaffolded to actually implement it

which brings us a lot of

um and and entails many codes of paint

and a lot of like Hands-On and working

with some of these questions that are

that we may have

surely the agent is a generous model

map territory

but then people sometimes do go beyond

and go into the ontological and and uh

but but that's that's definitely a

secondary

Claim about a system

one that might be merited or not but

it's a secondary claim so first

approximation we're just talking about

the modeling apparatus

that may describe something fundamental

in a given system

but also literally may not or just maybe

one of many things that

any other um

thoughts or questions people have

yeah well

a few more things from chapter three

and then into chapter four anyone who

has read it before Wanna Give a note on

chapter four

M Carl six

okay

any other questions or or we'll end

all right

all right thank you see you all next

time bye